Name: ________________R. No.___ Class/ Sec:_______ Date:__________ Invig. Sign_______

ATOMIC ENERGY CENTRAL SCHOOL, NARORA

CLASS XII COMPUTER SCIENCE HALF YEARLY EXAMINATION 2018-19

MM: 70 **TIME: 3 Hr.**

General Instructions:

- **All questions are compulsory and marks are mentioned in front of each question.**
- **Electronic devices are prohibited to use in the examination.**
- **Use Blue or Black Pen only.**
- **Programming language : C++**

1. (a) Differentiate between formal arguments and actual arguments. Also give a suitable C++ code to illustrate both. 2

(b) Name the header files that shall be required for successful compilation of the following C++ program : 1

```
int main( )
    {       char a, name[50];
            a = getch( );
            delay(100);
            if(isdigit(a))
            cout<<"\n enter your name \n";
            gets(name);
            if(strcmpi(name,"CSC Student")==0)
            cout<<"\n U r Welcome\n";
            else
            cout<<"\n Sorry u are not allowed\n";
    }
```

(c) Rewrite the following program after removing the syntactical errors(if any). Underline each correction. 2

```
#include<iostream.h>
struct Screen
{ int C, R;}
Void ShowPoint(Screen P)
{
cout<<P.C, P.R<<endl;
 }
void main()
{
 Screen Point1; cin>>Point1->C>>R;
ShowPoint(Point1);
 Screen Point2= point1;
 C.Point1+= 2; }
```

(d) Find the output of the following program: 3

```
#include<iostream.h>
void Change(int  Arr[], int Count)
{       for(int  c=1 ; c<Count ; c++ )
```

```
            Arr[c-1] += Arr[c];
}
void main( )
{       int A[] = {7,8,9}, B[] = {50,60,70,80}, C[] = {1000,1300};
        Change(A,3);
        Change(B,4);
        Change(C,2);
        for(int i=0 ; i<3 ; i++ )
        cout<<A[i]<<"$"<<endl;
        for(i=0 ; i<4 ; i++ )
        cout<<B[i]<<"$"<<endl;
        for(i=0 ; i<2 ; i++ )
        cout<<C[i]<<"$"<<endl;
}
```

(e) Find the output of the following program: 2

```
#include<iostream.h>
#include<conio.h>
        void main()
        {
                char a[]= "Computer 2018";
                int i;
                for(i=0; a[i]!= '\0';i++)
                {
                        if(a[i]>= 97 && a[i]<=122)
                                a[i] ++;
                        else
                        if(a[i]>= '0' && a[i]<= '9')
                                a[i] = a[i -1];
                        else
                        if(a[i]>= 'A' && a[i]<= 'Z')
                                a[i]+ = 32;
                        else
                                a[i]= '%';
                }
                        puts(a);
        }
```

(f) Study the following program and select the possible output from it: 2

```
#include<iostream.h>
#include<stdlib.h>
#define Success (N)        ((N%2 ==0)? N+1:N+2)
        void main( )
{               randomize( );
                        int num= random(3)+3;
                        for(int I= num ; I<=num+2;I++)
                        cout<<Success(I)<<'@';
}
```

Options: a) 3@5@7@
b) 7@7@9@
c) 7@9@9@
d) 7@9@11@

2.(a) Differentiate between Data Abstraction and Data Hiding with the help of an example in C++. 2

(b) Answer the questions (i) and (ii) after going through the following class: 2

```
class Pet	{
		public:
			char category[20];
			float cost;
			Pet( char xname[])		// function1
			{
				strcpy(category, xname)
			}
			Pet(float f)			// function 2
			{
				Cost = f;

			}
			Pet(Pet&t);		//function 3
		};
```

(i) Create an object, such that it invokes function1.
(ii) Write complete definition for function 3.

(c) Define a class Test in C++ with the following specifications : 4

Private Members
i) TestCode of type integer
ii) Description of type string
iii) NoCandidate of type integer
iv) CenterReqd (number of centers required) of type integer
v) A member function CALCNTR( ) to calculate the number of centers as (NoCandidate/500+1)

Public Members
i) A function SCHEDULE( ) to allow user to enter values for TestCode, Description, NoCandidate& call function CALCNTR( ) to calculate the number of centres.
ii) A function DISPTEST( ) to allow user to view the content of all the data members.

(d) Consider the following declarations and answer the questions given below: 4

```
class NATION
{
		int H;
	protected:
		int S;
	public:
		void INPUT();
		void OUTPUT();
};
class WORLD: private NATION
{
		int T;
	protected:
```

```
        int U;
    public:
        void  INDATA(int, int);
        void  OUTDATA();
};
class STATE: public WORLD
{
        int  M;
    public:
        void DISPLAY(void);
};
```
(i) Name the base class and derived class of the class STATE.
(ii) Name the data member(s) that can accessed from function DISPLAY ().
(iii)Name the member function(s) which can be accessed from the objects of class WORLD.
(iv) List all the members accessible by the objects of the class STATE.

3. (a) Define a function named search() which receives an array of integers and the size of the array as parameter. The function swaps first element of the array with the last element and second element with the second last element and so on. 3
For example
If the received array elements are 15,13,10,12,14,16,22
Then the rearranged array should be 22,16,14,12,10,13,15

(b) An array MAT[20[10] is stored in the memory along the column with each element occupying 4 bytes of memory. Find out the base address and the address of element MAT[13][7], if the location of MAT[12][5] is 2500. 3

(c) Write functions in C++ to add a node to a dynamically allocated Queue which contains city's information using the given structure. 4
```
struct  city
{
        char name[100];
        long  population;
        city * front , * rear ;
};
```
(d) Write a user defined function int ALTERSUM(int B[], int N, int M) in C++ find and return thesum of all alternate elements of a 2-D array starting from first element i.e.B[0] [0]. 2
EXAMPLE:

5	4	3
6	7	8
1	2	9

The function should add 5,3,7,1,9 and return 25.

(e) Evaluate the following postfix expression using a stack and show the contents of the stack Aftereach operation. 2
100, 40, 8, +, 20, 10, -, +, *

4. (a) Observe the program segment given below carefully and answer the question that follows: 1
```
class Team
{       long tid;
        chart name[20];
```

```
        float points;
    public :
        void  Accept();
        void  Show();
        void  PointChange(); // Function to change points.
        long  R_tid{ return tid; }
    };
    void Replace_points( long Id )
    {fstream file;
    file.open("TEAM.DAT", ios::binary | ios::in | ios::out );
    Team T;
    int Record = 0, Found = 0;
     long  int  pos = file.tellg();
    while(!found &&file.read((char*)&T, sizeof(T)))
     {
        if (Id == T.R_tid())
        {
            cout<<" Enter new points ";
            T.Pointchange();
    --------------------------------------------- //  statement 1
    --------------------------------------------- //  statement 2
            found = 1;
        }
        Record++ ;
        pos = file.tellg();
     }
        if(Found == 1) cout<<" Record updated ";
    file.close( );
    }
```

Write C++ statements for the statement 1 to position the write pointer at the beginning of the record for which the Team's Id matches with the argument passed andstatement 2 to write the updated record at that position.

(b) Write a function in C++ to count and display the no of times words "computer" and "science" occurs in a text file. 2

Example:

If the file contains:

Computer Science is the need of the hour and it must be learnt by all the science students.

Then the output should be: 3

(c)Given a binary file COMPUTER.DAT, containing records of the following type: 3

```
class Consumer
  { charc_name[20];
    int pbox;
    char area[20];
   public:
        void inconsumer( );         // to get the details of consumer.
```

```
        void outconsumer( );      // to display the details of consumer.
        int checkarea(char  ac[])  // checking area with the the value ac ,passed to it
        {
                return strcmpi(area,ac);
        }
};
```

Write a function that reads binary file **CONSUMER.DAT** containing records of above class, and displays all those records whose area is same as given by user.

5.(a) Convert the following infix expression to postfix using stack. Also show the contents of stack at each step. 2

A + B ^ C * D) * (E + F / D)

(b) Write a function arrange() in C++ to sort an array of structure Game in ascending order of points. 2

Note : Assume the following definition of structure Game.

```
struct sports
{       int points;
        char Teamname[20];
};
```

(c) Write a user defined function Diagonals() which receives a two dimensional array A and its size as argument. The function displays sum of all the elements on both the diagonals. 2

Example : If one 2D array elements are

2	3	1	5
0	3	1	5
2	7	1	5
2	4	0	5

Then the output should be

Sum = 26 (2+3+1+5+5+1+7+2)

6. (a) Write a function in C++ to display object from the binary file "PRODUCT.Dat" whose product price is more than Rs 200. Assuming that binary file is containing the objects of the following class: 3

```
class PRODUCT
{
        int PRODUCT_no;
        char PRODUCT_name[20];
        float PRODUCT_price;
 public:
        void enter( )
        {
                cin>>PRODUCT_no ; gets(PRODUCT_name) ;
                cin>>PRODUCT_price;
        }
 void display()
        {
                cout<<PRODUCT_no ; cout<<PRODUCT_name ;cout<<PRODUCT_price;
        }
 intret_Price( )
        {
                return PRODUCT_price;
        }
```

(b) Define a class Student with the following information: 4

Private members

a. rollno of type int
b. Name of type character with size as 50
c. Category of type string ("General" , " Schedule caste" , " Schedule Tribe", "Other Backward Cast")
d. Fees of type Float
e. A function Calfee() to calculate fees of the student on the basis of category entered as :
    a. Category 'General' or 'Other Backward Cast' then Fees is 5500
    b. Category 'Schedule caste' then Fees is 900
    c. Category 'Schedule Tribe' then Fees is 100

Public members

- A constructor to initialize category as a NULL and fees as 0.
- A function to input Roll number, name and category of the student, and call function Calfee() to calculate fees of the student.
- Function to display all the information of the student

(c)Each node of a stack contains the following information in addition to pointer field: 4

(i) Pin code of city
(ii) Name of the School

Give the structure of the node for the linked stack in question. Top is a pointer that points to the topmost node of the stack. Write a function ADD() To push a node into the stack which is allocated dynamically.

(7) (a)Differentiate between Global variable and local variable. Also give an example of each in C++. 2

(b) Explain encapsulation. 2

(c) An array MAT[10][20] is stored in the memory along row with each element occupying 8 bytes of memory. Find out the base address and address of element MAT[5][10] , if MAT[8][4] is stored at the address 1000. 3

(d) Explain Multiple inheritance with the help of an example. 2

****